Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *ARABICMMLU* | ***Context:*** *Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark measures performance as percentage MCQ accuracy [Q3, Q45] via highest-token-probability scoring for open-source models [Q50, Q79] or first-token regex matching for closed-source models [Q53, Q54]. This output form is fundamentally misaligned with the deployment's requirement for open-ended, explanatory, multi-perspective natural-language responses [elicitation A3]. The benchmark cannot evaluate whether a model produces appropriately nuanced, perspective-flagging answers — it can only score whether the model selects the curriculum-designated correct option. Additional output-form analyses (calibration [Q80], length-confidence correlation [Q82], few-shot trends [Q75, Q76], negation sensitivity [Q83, Q84]) all operate on the same MCQ scoring surface. The text-based MCQ form is appropriate for literacy-capable populations [Q93] in principle, but the form mismatch with the deployment's natural-language explanatory output requirement is structural and not addressable through reweighting or filtering. Supplementary frameworks (AraGen 3C3H, 'Beyond MCQ') exist [WEB-5, WEB-6] but are separate evaluations.

ID	Evidence
Q3	“Our comprehensive evaluations of 35 models reveal substantial room for improvement, particularly among the best open-source models.” (p.1)
Q45	“Our experiments focus on zero-shot and few-shot settings across 35 models.” (p.5)
Q50	“Following previous studies (Koto et al., 2023; Li et al., 2023), for open-source models, we determine the answer based on the highest probability among all possible options.” (p.6)
Q53	“For closed-source models, we determine the answer based on the first token generated in the text using a regular expression.” (p.6)
Q54	“If there is no match, we assign a random answer.” (p.6)
Q75	“While all the results in Section 4.2 were based on zero-shot learning, we observe in Figure 7 that when we move to few-shot learning, results for base models improve but those for instruction-tuned models deteriorate.” (p.8)
Q76	“Specifically, AceGPT and Jais show an improvement of 2–10 points when using few-shot learning, but the results for BLOOMZ and Jais-chat drop.” (p.8)
Q79	“The answer probability is obtained through softmax normalization across the available candidate answers.” (p.8)
Q80	“In Figure 8, we observe that the three open-source models are well calibrated with correlation scores r > 0.9.” (p.8)
Q82	“We find no correlation between the length of the questions and the model confidence for either Jais or AceGPT.” (p.8)
Q83	“Despite negation being an absolutely foundational linguistic phenomenon, LLMs have been shown to be worryingly insensitive to its effects in English (Kassner and Schütze, 2020; Hosseini et al., 2021; Truong et al., 2023).” (p.8)
Q84	“We thus perform an analysis of LLM performance over questions in ArabicMMLU with and without negation to determine whether this observation ports across to Arabic.” (p.8)
Q93	“ArabicMMLU is focused solely on text-based assessment, and the exploration of multimodal questions is left for future work.” (p.9)
WEB-5	'Beyond MCQ' Arabic open-ended benchmark — supplementary alternative for OF (https://arxiv.org/abs/2510.24328)
WEB-6	AraGen 3C3H LLM-as-judge framework for open-ended Arabic generation (https://arxiv.org/abs/2510.13430)

Strengths

- Standardized accuracy metric is reproducible and comparable across 35 evaluated models [Q3, Q45]

- Auxiliary analyses (calibration, length-confidence, negation sensitivity) provide robustness signal beyond raw accuracy [Q80, Q82, Q83]
- Two scoring methods documented for open-source vs closed-source models [Q50, Q53]

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Q80	“In Figure 8, we observe that the three open-source models are well calibrated with correlation scores r > 0.9.” (p.8)
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Mismatch — benchmark output is discrete MCQ labels [Q3, Q50, Q53]; deployment requires open-ended natural-language responses with multi-perspective acknowledgment [elicitation A3, output_form_mismatch].

OF-2: Not applicable — deployment is text-only [A4], so TTS availability is not required.

OF-3: Target population is literate non-native Arabic learners [target_population_description]; text output is appropriate, but the benchmark's discrete-label form does not assess natural-language fluency, explanatory quality, or perspective-flagging.

OF-4: Documented: external validity violation — benchmark MCQ accuracy will not predict whether the system produces appropriately explanatory, multi-perspective natural-language responses required by the deployment [elicitation A3, output_form_mismatch.mismatch_severity=HIGH].

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Q53	“For closed-source models, we determine the answer based on the first token generated in the text using a regular expression.” (p.6)
Q93	“ArabicMMLU is focused solely on text-based assessment, and the exploration of multimodal questions is left for future work.” (p.9)
Q94	“It is important to emphasize that our experimental results do not provide conclusive answers regarding the performance of LLMs in Arabic.” (p.9)

Remediation

Gap 1: MCQ accuracy scoring cannot evaluate explanatory natural-language quality required by deployment [Q3, Q50, Q53]

Recommendation: Adopt LLM-as-judge or human-rater scoring of free-form responses to a curated subset of ArabicMMLU history/civics/Islamic-studies questions reformulated as open-ended questions, scoring on correctness, completeness, and multi-perspective coverage.

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